

Support-Two Normal Forms for a Shared-Tag Pauli Compilation Grammar

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Abstract

Tag-and-Restore compilation couples auxiliary anticommuting Pauli frames, a shared label operator, target assignments, and factored corrections. We analyze a fixed six-target, three-block grammar under an explicit logical support-count objective. Every feasible solution can be transformed without increasing cost so that each auxiliary frame operator has support at most two. Support one is not uniformly sufficient: an exact two-qubit instance has unrestricted cost 5 and support-one cost 6. The sharp threshold yields a direct exact word-RAM algorithm with $O(n^9)$ time and $O(n^3)$ working memory for this fixed grammar. Standalone finite checks corroborate the two local proof lemmas and recompute the displayed sharpness instance. A pre-specified runtime comparison is adverse: the direct support-two implementation timed out on all six full-subject cells under a 120-second limit, although both solvers agreed on every jointly completed cell. The result is therefore a grammar-specific normal form and algorithmic upper bound, not evidence of faster compilation, hardware benefit, fault-tolerant resource advantage, or general block-encoding optimality.

Keywords: quantum compilation; Pauli operators; block encoding; exact optimization; normal forms; support complexity.

1 Introduction

Exact compilation is usually posed as optimization over a sufficiently expressive search space. A complementary question is whether that space admits a smaller exact normal form. Such a result can separate representational necessity from implementation performance and can expose the precise obstruction to a stronger restriction.

We study this question for a fixed specialization of a Tag-and-Restore block-encoding method [1]. That method explicitly leaves its anticommuting frames, controls, and compatible label operators open to user choice. We arrange six nonidentity Pauli targets into three ordered blocks, use one frame pair per block, and share one label operator. The objective counts auxiliary support and factored correction support with fixed coefficients. It is an analysis coordinate defined for this specialization, not the underlying method’s universal objective or a circuit-depth, fault-tolerant, or hardware cost model.

Main results and assumptions. All statements use exactly this six-target, three-block grammar and objective. First, we prove that every optimum has a representative in which each frame Pauli has support at most two. Second, an exact two-qubit counterexample proves that support one is not uniformly sufficient, so the uniform frame-support threshold is exactly two. Third, we count the ordered anticommuting support-two pairs and obtain a direct exact $O(n^9)$ word-RAM algorithm. Fourth, finite checks corroborate the local lemmas and exact implementation. Finally,

a pre-specified runtime study fails to show implementation benefit: the direct algorithm times out on all six full-subject comparisons.

The distinction between theorem and performance evidence is central. The normal form is an exact mathematical statement. The asymptotic result is an upper bound for a new direct algorithm, not a lower bound or a speed claim. The runtime study is retained as adverse evidence rather than converted into a positive result. No statement here implies physical-resource advantage or optimality for another grammar, objective, or compiler.

2 Grammar and objective

2.1 Pauli representation and block structure

Global phases are irrelevant to support counting. At each qubit a Pauli letter belongs to $\{I, X, Y, Z\}$, and the binary symplectic product $\langle \cdot, \cdot \rangle$ determines commutation. Fix blocks $b \in \{0, 1, 2\}$ and branches $j \in \{0, 1\}$. The six input targets are nonidentity phase-ignored Paulis P_{bj} . Their indices specify both the block pairing and the assignment of a target to a branch. If six unpaired targets are supplied instead, the 15 possible pairings form a constant-size outer family.

A feasible configuration chooses six nonidentity frame Paulis A_{bj} , a shared Pauli S , and one central branch c_b per block. The frame pair in each block must anticommute,

$$\langle A_{b0}, A_{b1} \rangle = 1,$$

and the shared operator must give every ordered pair the same syndrome

$$(\langle S, A_{b0} \rangle, \langle S, A_{b1} \rangle) = \lambda, \quad \lambda \in \{(1, 0), (0, 1)\}.$$

The nonzero syndrome bit makes S nonidentity automatically. These conditions, together with the target pairing and branch assignment, define the admitted family; there are no additional feasibility constraints.

For a target letter p and assigned frame letter f , the local correction letter is the phase-ignored product pf . Corrections in corresponding branches of the three blocks are factored together. For local letters a, b, c , the factor cost is

$$F(a, b, c) = w(a) + w(b) + w(c) - 2 \mathbf{1}[a = b = c \neq I],$$

where $w(I) = 0$ and $w(X) = w(Y) = w(Z) = 1$. Thus three identical nonidentity correction letters receive a two-unit sharing discount.

2.2 Declared objective

Let $m_{bj} = 2$ when $j = c_b$ and $m_{bj} = 4$ otherwise. The normalized objective is

$$C = \sum_{b=0}^2 \sum_{j=0}^1 m_{bj} (w(A_{bj}) - 1) + 2w(S) + \sum_q \sum_{j=0}^1 F(P_{0j,q} A_{0j,q}, P_{1j,q} A_{1j,q}, P_{2j,q} A_{2j,q}).$$

The subtraction of one support unit from each nonidentity frame removes the fixed raw-cost baseline $\sum_{b,j} m_{bj} = 18$. It leaves every optimizer unchanged and fixes the absolute cost convention used below. Under this normalization, removing one supported coordinate from a central frame refunds two units, while removing one from a noncentral frame refunds four. The shared operator costs

two units per supported qubit, and the last term factors the three corrections in corresponding branches.

For a fixed pairing, three target-order bits would give eight within-block orders. Simultaneously swapping the two branches in all blocks, relabelling the central branches, and replacing λ by its other orientation preserves feasibility and cost. We may therefore fix the order in block 0 and enumerate the four relative orders in blocks 1 and 2. For fixed frames, the correction and shared-operator terms do not depend on the central choices. Each block is minimized by assigning multiplier 2 to the heavier frame, with either choice allowed on a tie. Finally, no objective term other than $2w(S)$ depends on the compatible shared operator, so a minimum-weight compatible S suffices.

The unrestricted family contains every feasible configuration defined above. The support- k family adds only the restrictions $w(A_{bj}) \leq k$ for all b, j . Family containment gives one inequality between their optima; the reverse inequality follows from the transformation theorem.

2.3 Analytic and finite evidence

The mathematical proof uses the objective above, a parity-preserving exchange, and a well-founded support-reduction argument. Finite enumeration checks the local lemmas but is not the logical basis of the all-size theorem. The exact counterexample is a lower witness for support two. The included sharpness enumeration establishes only its stated two-qubit minimum. Runtime measurements address one implementation and panel, not the theorem's correctness.

3 Normal form, sharpness, and direct algorithm

3.1 A local correction bound

Lemma 1 (Correction exchange). *Fix one local target letter p , a nonidentity frame letter f , and the other two letters u, v in the same three-way factor. Replacing pf by p increases the factor cost by at most two:*

$$F(p, u, v) - F(pf, u, v) \leq 2.$$

Proof. Only one letter changes, so the ordinary nonidentity count rises by at most one. If the old triple did not receive the sharing discount, a new discount can only reduce the change. If the old triple did receive it, then pf was nonidentity and equal to $u = v$. Destroying the discount adds two, but the changed letter cannot increase the nonidentity count: either $p = I$ and the count falls by one, or p is another nonidentity letter and the count is unchanged. Hence the increase is at most two. $\square$

3.2 A parity-preserving subset

For one frame A , its anticommuting partner A' , and the shared operator S , assign each supported coordinate q the class

$$c_q = (\langle A_q, A'_q \rangle, \langle S_q, A_q \rangle) \in \mathbb{F}_2^2.$$

Global anticommutation implies that the first components sum to one.

Lemma 2 (Zero-sum subset). *If $w(A) \geq 3$, there is a nonempty proper subset $Q \subset \text{supp}(A)$ with $|Q| \leq 2$ and $\sum_{q \in Q} c_q = (0, 0)$.*

Proof. A zero class supplies a singleton. Otherwise, a repeated nonzero class supplies an equal pair, whose sum is zero. If neither occurs, all classes are distinct and nonzero. There are only three

nonzero classes, so support at least three forces the support-three multiset containing each nonzero class once. Its first components sum to zero, contradicting global anticommutation. $\square$

3.3 Exact support threshold

Theorem 1 (Cost-nonincreasing support-two transformation). *For every admitted instance and every feasible configuration, there is a feasible configuration of no greater cost in which every frame Pauli has support at most two. Consequently, for every qubit count the unrestricted optimum equals the support-two optimum.*

Proof. Start from any feasible configuration. If a frame has support at least three, choose Q from the zero-sum lemma and replace its letters on Q by identities. The first parity condition preserves anticommutation with the partner, and the second preserves the shared-label syndrome. Because Q is proper, the modified frame remains nonidentity. At each removed coordinate the frame refund is at least two, while the correction-exchange lemma bounds the correction increase by two; the shared operator is unchanged. Cost therefore does not increase, and total frame support strictly decreases. Iteration terminates in the support-two family. Applying the transformation to an unrestricted optimum gives one optimum inequality, while family containment gives the other. $\square$

The threshold is sharp. Write two-qubit Pauli strings with the qubit-0 letter first and take the three target pairs

$$(XI, XI), \quad (XI, XI), \quad (IX, IY).$$

A feasible support-two solution uses frame pairs (YI, XI) , (YI, XI) , and (ZX, IY) , marks the first branch central in each block, and takes $S = XI$ with common syndrome $(1, 0)$. The normalized frame charge is 2, because only the central frame ZX has weight above one. The shared-operator charge is 2. On the first branch, all three corrections equal ZI , so their factor cost is 1; every correction on the second branch is the identity. The total is therefore $2 + 2 + 1 = 5$.

The support-one lower bound is exhaustive. On two qubits there are 12 ordered anticommuting support-one pairs per block. The enumerator covers their Cartesian cube, four relative target orders after the global branch-swap reduction, both syndrome orientations, and every minimum-weight compatible shared operator. Central assignments are optimized exactly by placing multiplier 2 on the heavier frame; with support-one frames all central choices tie. The resulting minimum is 6. Thus the uniform support threshold is exactly two.

3.4 Constructive pair count and complexity

There are $3n + 9\binom{n}{2}$ nonidentity phase-ignored Paulis of support one or two. For a weight-one first frame, the number of anticommuting support-two partners is $6n - 4$. For a weight-two first frame, splitting partners into weight one on the active support, weight two on the same support, and weight two with one shared coordinate gives $12n - 16$. The number of ordered anticommuting pairs is therefore

$$N_{\text{pair}}(n) = 3n(6n - 4) + 9\binom{n}{2}(12n - 16) = 54n^3 - 108n^2 + 60n.$$

The input consists of six explicit n -qubit target strings, so its representation has $\Theta(n)$ Pauli letters. One block pair uses at most three coordinates, and three pairs therefore use at most nine. A minimum-cost shared operator has no support outside this active union, since outside letters change no constraint and only add cost. Its six symplectic equations can therefore be solved by a

Table 1: Evidence layers and the inferences they support.

Evidence	Supported inference
Analytic exchange proof	Exact support-two normal form for the declared grammar and objective
Exact paired minima $5 < 6$	Support one is not uniformly sufficient
Constructive count	Direct $O(n^9)$ word-RAM upper bound
Finite checks	Corroboration of the local lemmas and displayed sharpness instance
120-attempt runtime study	No measured practical benefit; twelve timeouts retained

constant-size 64-syndrome dynamic program. After one linear scan of target coordinates, correction costs for each candidate are updated on at most nine active coordinates. Enumerating three ordered frame pairs yields

$$T(n) = O(n + N_{\text{pair}}(n)^3) = O(n^9)$$

on a word RAM, with $O(n + N_{\text{pair}}(n)) = O(n^3)$ working memory. A qubit index and local target letter occupy constant-size words in this model. Charging their bit operations and the accumulated cost explicitly adds polylogarithmic factors. The result is a constructive polynomial exact-solvability bound. No same-problem asymptotic improvement, lower bound, or exponent optimality is established.

3.5 Finite verification and adverse runtime result

A standalone implementation checks all 192 local correction cases and all odd-parity class tuples through support eight. It finds a maximum factor-cost increase of two, no subset-lemma failure above support two, and exactly four ordered support-two obstruction patterns. A second packaged check exhaustively obtains support-two minimum 5 and support-one minimum 6 for the displayed paired instance. These finite checks corroborate the analytic argument and sharpness instance; they do not prove the all-size theorem or provide external replication.

The implementation-performance study is adverse. Its two recorded subjects were evaluated on one-, two-, and three-qubit prefix projections and on their original eight- or twelve-qubit strings, called the full-subject cells. Of 120 attempts specified before measurement, 108 completed and 12 timed out. Both exact solvers agreed on every jointly completed correctness cell, and every completed witness verified. The direct support-two implementation timed out in all six three-qubit scale cells and all six full-subject cells under the 120-second limit. The unrestricted solver completed every corresponding cell. Consequently no runtime or memory improvement is established.

4 Discussion

The theorem separates representation from performance. It removes arbitrary-support frames from exact optimization without asserting that a direct enumeration is practically efficient. The adverse runtime study illustrates why this distinction matters: a polynomial exponent and a smaller normal form do not automatically outperform a factorized unrestricted dynamic program on finite workloads.

The sharpness witness also identifies the obstruction to support one. At support two, no nonempty proper subset need preserve both frame anticommutation and the shared-label syndrome.

A locally removable frame letter can therefore be globally coupled to the shared operator. The upper proof and lower witness meet at this parity boundary rather than at an empirical cutoff.

The result is useful primarily as a formalization of one cost model. The underlying method already treats its design choices as constraint-dependent, so objective dependence is expected rather than a contradiction of that construction. The present analysis supplies a complete representational reduction, an exact algorithmic upper bound, and an auditable negative performance result. It does not establish a new universal synthesis primitive. Practical significance would require either a better algorithm exploiting the normal form, a broader theorem, or an independently evaluated resource improvement under a physically meaningful model.

5 Related work

The studied grammar descends from version 4 of the Tag-and-Restore block-encoding preprint [1]. Its Section 4 introduces the primitive; Theorem 1 and Remark 1 leave the anticommuting frames, ancilla count, and controls open to user choice; and Theorem 2 and Remark 2 expose nonunique, optimizable label operators. Its numerical study fixes one canonical frame family and optimizes label strings independently. We inherit the primitive and those design freedoms. The present analysis fixes six target slots in three paired blocks, branchwise three-way factorization, one common binary syndrome, the two/four frame multipliers, and the normalized logical objective. The support restriction is an analysis family; the cost-nonincreasing transformation, sharpness instance, and induced direct bound concern this specialization only. This division identifies the scientific objects but makes no priority claim.

Anticommuting-unitary partitioning groups anticommuting Pauli terms for measurement, which is a different task and optimized quantity [2]. Pauli-cluster and Pauli-level compilers transform circuits or Hamiltonians in Pauli representations; they do not establish a same-object performance improvement for the grammar studied here [3, 4, 5, 6].

Support reduction itself is not new as a general idea. Phase-gadget synthesis decomposes Pauli exponentials using one- and two-qubit components [7], and parity-network work exploits sharing across related terms [8]. Two-qubit gates are universal for quantum computation [9], while stabilizer-state normal forms are classical tools of the Pauli formalism [10]. These results remove any claim that support-two building blocks, parity sharing, or two-local universality originate here. They address different objects from the grammar-specific statement proved above: the present theorem transforms auxiliary frames while preserving one fixed shared-label objective and exact optimum. Hamiltonian weight reduction and perturbative gadgets likewise change the encoded Hamiltonian, often by adding degrees of freedom or approximation, rather than proving this cost-monotone frame exchange [11, 12].

Bounded web and arXiv searches were refreshed through 28 August 2026 using query families for Tag-and-Restore encoding, shared labels, Pauli frames, support-two normal forms, exact expressivity, binary-symplectic compilers, and sharp support thresholds. They also inspected exact and approximate synthesis, low-ancilla block encodings, fault-tolerant resource models, and compiler benchmarks. No exact equivalent of the full grammar-specific theorem was located, but this search cannot establish absence or priority. The residual contribution is therefore stated descriptively: a cost-monotone normal-form proof, its sharp obstruction, and the induced exact upper bound for this declared grammar. Novelty beyond that narrow formulation remains a matter for independent expert review.

6 Limitations

The theorem is scoped to six targets, three blocks, one shared binary label, the stated factor rule, and fixed support multipliers. Changing the objective can make support three beneficial. Other term counts, shared-label ranks, block grammars, routing constraints, or factor rules require new proofs.

The direct complexity result is a word-RAM upper bound. It is not a lower bound, an exponent-optimality result, or an asymptotic statement about other implementations. The finite sharpness check uses the included solver’s feasibility and witness routines, so it is corroboration rather than an externally independent implementation.

The runtime panel is small and adverse. It provides no acceleration, memory, energy, logical-depth, fault-tolerant, or hardware claim. Earlier finite chemistry observations and heuristic structural-cost reductions are not used as theorem or practical-value evidence here. Patents, private manuscripts, unindexed software, post-search publications, and independent venue-level novelty judgments were not assessable in the bounded literature audit.

7 Conclusion

For the declared shared-Tag Pauli compilation grammar, every optimum has a support-two frame representation, and an exact two-qubit witness shows that support one is insufficient. The sharp normal form yields a direct exact $O(n^9)$ word-RAM algorithm. Finite checks corroborate the local lemmas and sharpness calculation, but a pre-specified runtime comparison finds no practical advantage because the direct implementation times out on every full-subject cell. The durable result is therefore an exact, objective-scoped normal form with a preserved negative performance outcome, not a claim of general compiler or physical-resource superiority.

8 Data and code availability

The manuscript source, standalone finite checker, direct solver, sharpness enumeration, runtime specification, anonymized row-level observations, recorded environment, deterministic aggregation check, and adverse result are available in the accompanying anonymous review archive. The archive includes integrity and verification instructions but no author metadata or protected data. Protected data are not required for any claim in this paper.

Verification should proceed in three layers: rerun the standalone local-lemma checker; recompute the sharpness witness and support-one minimum; then regenerate the runtime summary from all 120 retained rows without changing the time limit, panel, or success rule. The twelve timeout rows are part of the result and must not be discarded. The review archive supports deterministic audit of those recorded measurements, but does not contain the full original unrestricted measurement stack for a new timing campaign.

9 Ethics, author contributions, and resource statement

This work uses no human participants, animals, clinical data, or personal data. Its principal integrity risks are overgeneralizing a grammar-specific theorem and suppressing negative performance evidence; the manuscript addresses both by stating the exact scope and retaining every timeout.

The author is responsible for the mathematical claims, source verification, code, and final manuscript. Large language models were used for literature-search assistance, proof-audit prompts, code and prose refinement, and manuscript formatting. Mathematical statements, citations, calculations, and interpretive decisions were checked against the cited sources and underlying evidence during preparation; the author retains final responsibility for the manuscript.

No energy or carbon estimate is claimed because the records do not support a reliable life-cycle comparison. The finite checks and runtime study used ordinary CPUs.

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